

```
$ node breakfast.js --mode=async
```

JavaScript □□□

```
// 00000000 Promise 0 async/await
```

sync

□□□□

promise

□□□□□

await

□□□□□

event loop

□□□□

\$ cat menu.md



01 /



vs



02 / Promise

then/catch

03 / async & await

vs

04 / Event Loop

VIP

async

PART 01



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```
1  function makeCoffee() {
2      console.log("☕☕☕☕...");
3      const start = Date.now();
4      while (Date.now() - start < 2000) {} // ☐☐☐
5      console.log("☕☕☕☕");
6  }
7
8  function prepareBreakfastSync() {
9      makeCoffee(); // 2s
10     makeToast(); // 1.5s
11     fryEgg();     // 1s
12 }
```

CONSOLE — 4.5s

☕☕☕☕...

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```

CONSOLE - ~2s

0000...
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0000000000000000
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```

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\$ node breakfast.js

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☕ vs 🍞□□□□□□□□

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▶ 🍞🍞🍞🍞🍞🍞🍞🍞

☕ ☕☕☕2s□

🍞 ☕☕☕1.5s□

🔍 ☕☕1s□

☕☕☕☕ = 2 + 1.5 + 1 = 4.5 ☕☕ ☕☕☕☕☕ = max(2, 1.5, 1) = 2 □

PART 02

Promise

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□□□□□□□□

pending

□□□□□□

fulfilled

□□□□resolve("🍌🍌")

rejected

□□□□reject("🍌🍌")

```
1 let myPromise = new Promise((resolve, reject) => {
2   let success = true;
3   setTimeout(() => {
4     if (success) resolve("🍌🍌");
5     else reject("🍌🍌");
6   }, 2000);
7 });
8
9 myPromise.then(result => console.log(result))
10 .catch(error => console.log(error));
```

□□□□□ Promise.race + AbortController

Promise.race

HdPbO
HdPbO
HdPbO
HdPbO
HdPbO
HdPbO
HdPbO
HdPbO
HdPbO
HdPbO

AbortController

PART 03

async / await

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then **NOGRAPH** → **await**

```
await  then  async/await  then
```



```
1  async function serial() {  
2    const coffee = await makeCoffee();  
3    const toast = await makeToast();  
4    return [coffee, toast];  
5  }
```

TOTAL TIME

$$\approx 2 + 1.5 = 3.5s$$



token API —



```
1  async function parallel() {  
2    const [coffee, toast] = await Promise.all([  
3      makeCoffee(),  
4      makeToast(),  
5    ]);  
6    return [coffee, toast];  
7  }
```

TOTAL TIME

$$\approx \max(2, 1.5) = 2s$$



API

Promise.all vs allSettled

	Promise.all	Promise.allSettled
□□□□□	 reject □□□□	□□□□□□□
□□	□□□□	{status, value/reason} □□
□□	□□□□□□□	□□□□□□□□□□

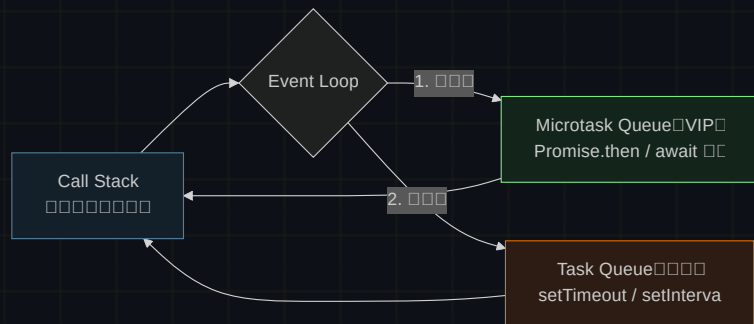
\$ npm run all □□□□□□□□□□allSettled □□□□□□□

PART 04

Event Loop

0000 VIP 0000000000

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□ □ □ □ **VIP** □
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```
1 console.log("1");
2 setTimeout(() => console.log("2"), 0);
3 Promise.resolve().then(() => console.log("3"));
4 console.log("4");
```

ANSWER

1 → 4 → Call Stack

→ 3 → VIP

→ 2 → 0

1 → 4 → 3 → 2

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□□□□□pending → fulfilled / rejected

await Promise.all

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Breakfast served.

fetch AbortController

```
$ node breakfast.js --mode=async
```

promise → await → race → event loop